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# Improving the Well-Being of Elderly Patients via Community Pharmacy-Based Provision of Pharmaceutical Care

## A Multicentre Study in Seven European Countries

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## Abstract

**Objective:** This study aimed to measure the outcomes of a harmonised, structured pharmaceutical care programme provided to elderly patients (≥65 years of age) by community pharmacists in a multicentre international study performed in 7 European countries.

**Design and setting:** The study was a randomised, controlled, longitudinal, clinical trial with repeated measures performed over an 18-month period. A total of 104 intervention and 86 control pharmacy sites participated in the research and 1290 intervention patients and 1164 control patients were recruited into the study.

**Main outcome measures and results:** A general decline in health-related quality of life over time was observed in the pooled data; however, significant improvements were achieved in patients involved in the pharmaceutical care programme in some countries. Intervention patients reported better control of their medical conditions as a result of the study and cost savings associated with pharmaceutical care provision were observed in most countries. The new structured service was well accepted by intervention patients and patient satisfaction with the services improved during the study. The pharmacists involved in providing pharmaceutical care had a positive opinion on the new approach, as did the majority of general practitioners surveyed. The positive effects appear to have been achieved via social and psychosocial aspects of the intervention, such as the increased support provided by community pharmacists, rather than via biomedical mechanisms.

**Conclusions:** This study is the first large-scale, multicentre study to investigate the effects of pharmaceutical care provision by community pharmacists to elderly patients. Future research methodology and implementation will be informed by the experience gained from this challenging trial.

In recent years, much attention has focused on the provision of pharmaceutical care, with many pharmaceutical organisations, institutions and pharmacists across the world adopting pharmaceutical care as the goal for pharmacy practice.[1] Pharmaceutical care has been defined as the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient's quality of life.[2] Such outcomes should include clinical, economic and humanistic aspects.[3,4] In the primary healthcare setting, pharmaceutical care encompasses community pharmacists being involved in designing, implementing and monitoring therapeutic plans in collaboration with other healthcare professionals, which can be integrated with all other aspects of care, e.g. medical care and nursing care.

Prior to the widespread implementation of pharmaceutical care, there is a need for large-scale, quality research into the impact of, and barriers to, pharmaceutical care provision in community pharmacy.[4-7] Some studies have reported the tangible benefits of pharmaceutical care provision to patients with specific disease states,[7] for example, pharmaceutical care programmes provided to patients with asthma,[8,9] diabetes mellitus[10] and hypertension.[11] More generic research projects have also reported improved patient outcomes such as improved quality of care and reduced medication costs,[12] improved appropriateness of prescribing in patients receiving polypharmacy[13] and decreased physician consultations.[14]

To date, there has been little research into the impact of community pharmacy-based pharmaceutical care provision for elderly patients.[6] Although it is encouraging that previous studies of community pharmacy-based pharmaceutical care provision for this patient group have reported that the model can be successfully implemented, evidence for the effect of pharmaceutical care on clinical, economic and humanistic outcomes in elderly patients has been limited.[15-18] The elderly are prescribed more drugs than younger people,[19] they often have a number of comorbidities[20,21] and they have an increased potential for drug-related problems.[22,23] Drug-related problems include inappropriate prescribing,[24-27] noncompliance with prescribed medications[28,29] and adverse drug reactions or interactions.[30,31] In addition, many geriatric hospital admissions are directly related to drug-related problems.[32-34] Therefore, there is major potential for the elderly to benefit from pharmaceutical care provision by community pharmacists as many drug-related problems and the detrimental aspects of suboptimal prescribing are preventable with careful patient assessment, patient education and patient monitoring. These are all core components of pharmaceutical care provision.

The potential economic and social benefits of such a targeted intervention programme are high, not only because of the potential savings resulting from decreased hospitalisation rates but also from the improved quality of life that can be achieved by elderly patients. Therefore, the aim of the present study was to investigate the impact of a coordinated community pharmacy-based pharmaceutical care programme for elderly patients on a range of health and economic outcomes. A further aim was to evaluate the views of patients, community pharmacists and general medical practitioners (GPs) on the novel mode of delivery of pharmaceutical services within the primary care setting.

## Patients and Methods

Centres in 7 European countries participated in the study: Denmark, Germany, The Netherlands, Northern Ireland (coordinating centre), Portugal, Republic of Ireland and Sweden. The methodology was developed following close collaboration between all of the European researchers involved in the multicentre study. The core methodological components used by each participating country are described here.

### Study Design

The study was a randomised, controlled, longitudinal, clinical trial with repeated measures being performed over an 18-month period. Ethical approval to perform the research was granted from relevant bodies within each of the participating centres, as dictated by local custom and practice. A number of interested community pharmacies were recruited within each country, with the pharmacy acting as the unit of randomisation (i.e. participating pharmacies were randomly assigned as control or intervention sites, with intervention pharmacies recruiting only intervention patients and control sites recruiting control patients).[35] The target sample size was 480 patients per country (half intervention, half control). This would enable a 5 to 10 point difference in all domains of the 36-Item Short Form health survey (SF-36) to be detected with 80% power (accounting for patients who withdrew).

#### *Study Sites*

Study sites were selected using the responses of community pharmacists who expressed interest in participating in the research, following publicity via mailshots, advertisements in pharmaceutical publications and at professional meetings. From these interested pharmacies, a minimum of 12 sites per country were chosen according to specific criteria set within each participating country relating to, for example, the population of elderly patients who visited the pharmacy, staffing levels within the pharmacy and working relationships with local GPs. Half of the recruited sites were randomly assigned as control sites and half as intervention sites and, where possible, control and intervention sites were matched as closely as possible according to size (e.g. total number of patients served), situation (e.g. city centre *vs* village) and type (e.g. owned by a single proprietor *vs* part of a national chain).

#### *Patients*

Eligible patients were 65 years or older, taking 4 or more prescribed medications and orientated with respect to self, time and place. In addition, they were community dwelling and regular visitors to a recruited community pharmacy. Patients were excluded if they were housebound or resident in a nursing/residential home. Identification of patients was performed via a personal approach by the pharmacist within the pharmacy, via GP records or from records kept within the pharmacy. Those patients who were willing to participate were recruited into the study and written consent was obtained prior to the commencement of data collection.

#### *Training of Intervention Pharmacists*

Pharmacists employed at intervention sites were trained to provide the structured pharmaceutical care programme to intervention patients, whilst control pharmacists continued to provide normal services to recruited patients. Intervention pharmacists attended at least one study day organised by each national research centre, where they received training on the implementation of the study and the provision of pharmaceutical care. To account for differences in education and the practice of pharmacy in the participating countries, a study manual was developed to harmonise the implementation of pharmaceutical care. Each intervention pharmacist received this extensive study manual, which was developed by the research group. The manual contained an overview of the concept of pharmaceutical care and its provision to elderly patients, detailed information on the therapeutic management of a number of disease states common in the elderly, together with other general issues pertinent to drug therapy in the elderly, for example, compliance and altered pharmacokinetics. Each country adapted the manual, translating and modifying sections where appropriate, according to differing national practices.

### Pharmaceutical Care Interventions

Prior to commencement of the study, community pharmacists were instructed to inform GPs within their locality of their participation in the study, providing them with details of the nature of the project and the pharmaceutical care that they would be providing. Where possible, formal links between the community pharmacists and GPs were established to encourage GP participation in, for example, the rationalisation of drug therapy.

As part of the pharmaceutical care provision, intervention pharmacists were asked to actively assess patients individually and to identify actual and potential drug-related problems using a structured approach. Possible drug-related problems included poor compliance, poor knowledge of drug therapy, adverse drug reactions/interactions and suboptimal drug regimens. Intervention pharmacists were encouraged to utilise a number of data sources in this assessment procedure, including some or all of the following: (i) the patient (via informal questioning); (ii) the patient's GP; and (iii) pharmacy-held records.

Following patient assessment, pharmacists were instructed to formulate (as necessary) an intervention and monitoring plan for each individual patient. Pharmacy interventions included: (i) educating the patient about their drug regimen and their medical condition(s); (ii) implementing compliance-improving strategies such as drug reminder charts; and (iii) rationalising and simplifying drug regimens in collaboration with the patient's GP. This third aspect of pharmaceutical care provision was structured using drug use profiles (graphical, chronological reviews of drug usage, originally devised at the University of Utrecht, Utrecht, The Netherlands, by a group led by Professor A. Porsius). This process of patient assessment and pharmacist intervention was a continuous process throughout the 18-month study.

### Outcome Measures and Data Collection

Data relating to health and economic outcomes were collected for all patients at baseline and at 6, 12 and 18 months; data on additional (intermediate) outcomes were also collected at each assessment point. Demographic data (i.e. age, gender, living arrangements, etc.) were collected at baseline and patients' orientation to self, time and place were determined using a series of questions. Both self-completed and interview-administered questionnaires were employed to collect data throughout the study. To minimise bias in data collection, the interviews were performed, where possible, by a member of staff other than the pharmacist, for example, by a pharmacy assistant. The methods of data collection described in this section were used by each participating country, with standard forms and questionnaires being developed for use during the study.

#### *Health Outcomes*

The health outcomes assessed included health-related quality of life, number of hospitalisations (i.e. admissions to hospital excluding emergency room visits), sign and symptom control and patient satisfaction with the services provided. Health-related quality of life was measured using the SF-36 health survey, a standardised, generic quality-of-life measure that assesses quality of life under 8 dimensions[36] that has been extensively validated in a number of different countries and languages.[37] Each country used the appropriately translated and validated SF-36 health survey, and the questionnaires were self-completed by the patients. The number of hospitalisations, sign and symptom control and patient satisfaction with the services provided were self-reported by patients within the questionnaires developed for use in the study by the international research group. These questionnaires were also self-completed by the patients.

#### *Economic Outcomes*

Comparisons between the 2 groups were made with respect to healthcare-related resource usage. The direct costs of the study were included in the analysis; indirect costs such as travel costs incurred by patients were not examined. The total cost of the intervention (i.e. provision of pharmaceutical care to the elderly) was divided into several cost components, which included: (i) cost associated with additional time spent by pharmacists; (ii) cost associated with contacts with GPs, specialists and nurses; and (iii) cost of hospitalisations and drugs. Data for the cost analysis were extracted from a number of sources: (i) pharmacist diary (pharmacist's time); (ii) self-completed questionnaire in which patients self-reported the use of healthcare resources (e.g. number of contacts with nurse, number of contacts with GP, number of admissions to hospital); and (iii) pharmacy-held medication records (cost of drugs prescribed). All costs were converted to euros (EUR) and were updated to January 1999 prices using the Retail Price Index. The costs used in the model were assumed to be representative of the actual costs that would be incurred in practice, with protocol-driven costs being ignored. It was not appropriate to make direct comparisons between centres because of the differing healthcare systems within each country.[38]

#### Intermediate Outcomes

The additional outcome measures assessed included: (i) patient knowledge of medicines; (ii) compliance with dosage regimens; (iii) number of contacts with GPs; (iv) prescription and nonprescription drug use; (v) number of times medication regimens were changed; (vi) acceptance by GPs of treatment recommendations made to them by the pharmacists; and (vii) professional satisfaction of pharmacists involved in the study and satisfaction of the GPs with the participation of the pharmacists in patient care.

Patient knowledge of medicines was assessed for correctness in 4 areas:

- medical indications for the drugs;
- number of dosage units taken per dose;
- number of doses taken per day;
- awareness of potential adverse effects.

An interview-based questionnaire was used to collect the data. A percentage score relating to the knowledge of all prescribed medicines was calculated for each patient, with higher scores indicating better knowledge.

The determination of compliance was based on the patients' responses to 4 items within a questionnaire designed for use in the study which was self-completed by all patients. The items reflected different aspects of noncompliance, namely forgetting doses, choosing not to take a dose because of feeling well, choosing not to take a dose because of perceived non-benefit and choosing to take more of a medicine than prescribed because of feeling the need for it, and were based on a previously validated scale.[39] Patients responded using a 4-point, Likert-type scale ranging from 'often' to 'never'. Patients were categorised as being compliant if they responded that they never experienced any of the aspects of noncompliance.

The number of contacts with their GP, prescription and nonprescription drug use and the number of times medication regimens were changed were self-reported by patients via self-completed questionnaires developed for use in this study.

Acceptance by GPs of the treatment recommendations made to them by pharmacists and professional satisfaction with the programme were assessed using the responses to questions in a self-completed questionnaire given to the appropriate healthcare professionals.

### Statistical Analysis

A uniform data format was defined for all information collected during the study and completed data were entered into one of the following data entry computer programmes: Epi-info Version 6 or SPSS for Windows®. All statistical analyses was performed using SPSS for Windows®. Data for all centres were pooled and the analysis was performed on the pooled data and on individual country data. Between-group comparisons (i.e. intervention and control) were assessed using independent t-tests where data were normally distributed, while the Mann-Whitney test was used where data were not normally distributed. Within-group analysis was performed using paired t-tests where the data were normally distributed, and the Wilcoxon test was used where the data were not normally distributed. Categorical data were assessed using Chi-squared analysis. To investigate changes over time, differences between each assessment period and baseline were computed for continuous data and comparisons between control and intervention patients were made using independent t-tests. Where appropriate, a summary measure was used to analyse continuous data, for example, SF-36 health survey domain scores. This was based on an area under the curve (AUC) calculation of the values reported at baseline and at 6, 12 and 18 months. For all statistical tests, the *a priori* value for significance was set at $p < 0.05$.

## Results

### Participating Pharmacy Sites and Recruited Patients

The 18-month study was completed in 5 of the 7 participating countries. Owing to operational difficulties, only 6 months' data were collected in the Republic of Ireland, and in Portugal the study was ended at 12 months instead of 18 months. In addition, the study lasted for 24 months in The Netherlands, with data being collected at baseline and at 6, 12 and 24 months (the 24-month data were treated as 18-month data in this case).

A total of 104 intervention sites and 86 control sites participated in the study. Table I displays the number of control and intervention sites recruited in each country and the number of patients recruited within each country. In total, 1290 intervention patients and 1164 control patients were recruited. Demographic details and baseline data for the pooled sample are displayed in table II. The control and intervention patients in the pooled data were well matched at baseline and there were no statistically significant differences between the 2 groups.

The number of patients assessed at each time period were as follows: 2454 at baseline (1290 intervention, 1164 control); 1977 at 6 months (1024, 953); 1627 at 12 months (863, 764); and 1340 at 18 months (704, 636). The reasons for drop-outs included patient withdrawal due to unwillingness to continue with the study, patients becoming too ill to participate or moving away from the area, pharmacy withdrawal due to lack of time or higher priority activities to be performed within the pharmacy, and patient death. Some countries had higher rates of drop-out than others, e.g, Northern Ireland (intervention 31.8% and control 56.7%) and The Netherlands (intervention 53.7% and control 47.1%); however, most of the patients who withdrew did so within the first 6 months. There were some significant differences between those patients who completed the study and those who were lost to follow-up. Those who withdrew from the study were significantly older (Mann-Whitney test; $p < 0.05$) and reported poorer quality of life at baseline (Mann-Whitney test; $p < 0.05$).

### Health Outcomes

#### *Health-Related Quality of Life*

There was a general trend for quality of life in all domains to decline over time in both the control

**Table I.** Numbers of pharmacy sites and patients by participating country

| Country | Intervention group | | | Control group | | |
|---|---|---|---|---|---|---|
| | no. of pharmacies | no. of patients | mean no. of patients per pharmacy | no. of pharmacies | no. of patients | mean no. of patients per pharmacy |
| Denmark | 14 | 254 | 18.1 | 14 | 269 | 19.2 |
| Germany | 35 | 193 | 5.5 | 13 | 102 | 7.8 |
| The Netherlands | 19 | 423 | 22.2 | 25 | 416 | 16.6 |
| Northern Ireland | 5 | 110 | 22.0 | 5 | 81 | 16.2 |
| Portugal | 10 | 54 | 5.4 | 13 | 86 | 6.6 |
| Republic of Ireland | 10 | 85 | 8.5 | 5 | 46 | 9.2 |
| Sweden | 11 | 171 | 15.5 | 11 | 164 | 14.9 |
| **Total** | 104 | 1290 | 12.4 | 86 | 1164 | 13.5 |

**Table II.** Patient demographics and baseline data. Data are presented as mean ± SD unless otherwise specified

| Variable | Intervention group (n = 1290) | Control group (n = 1164) |
|---|---|---|
| Age (y) [median ± interquartile range] | 74 ± 8 | 74 ± 8 |
| Gender (% male/% female) | 42.1/57.9 | 42.7/57.3 |
| Patients living alone (%) | 37.2 | 37.7 |
| Patients requiring help with daily activities (%) | 50.9 | 47.4 |
| Mean number of prescribed medicines | 7.1 ± 2.5 | 7.0 ± 2.5 |
| Patients using nonprescribed medicines (%) | 52.0 | 52.1 |
| Knowledge of medicines (%) | 65.1 ± 14.51 | 63.2 ± 17.0 |
| Health-related quality of life (SF-36) | | |
| general health | 46.2 ± 20.4 | 46.5 ± 21.0 |
| mental health | 69.7 ± 21.5 | 70.1 ± 21.6 |
| physical functioning | 46.6 ± 26.1 | 47.4 ± 27.4 |
| role emotional | 55.9 ± 44.6 | 56.2 ± 43.4 |
| role physical | 32.2 ± 40.4 | 32.9 ± 40.6 |
| vitality | 48.3 ± 22.9 | 49.6 ± 23.9 |
| bodily pain | 56.0 ± 27.8 | 55.4 ± 55.4 |
| social functioning | 67.4 ± 27.8 | 68.1 ± 29.0 |

**SD** = standard deviation; **SF-36** = 36-Item Short Form health survey.

and intervention patients (table III). In the pooled data, there were no significant differences between the control and intervention patients in any of the 8 dimensions over time (AUC summary measure analysed; independent t-test, $p > 0.05$). A number of significant differences in some domains were evident in the individual country analyses. In the Danish data, significant improvements in general health and role emotional scores were observed in intervention patients compared with control patients (independent t-test, $p < 0.05$). However, conflicting results were obtained from the data in Northern Ireland as scores for the physical functioning, role physical and vitality domains in the intervention group decreased significantly compared with an observed improvement in the control group (independent t-test, $p < 0.05$). These findings were, however, largely driven by patients attending one control site pharmacy who showed marked improvements in SF-36 health survey scores over time.

#### *Hospitalisations*

Data on hospitalisations of acceptable quality were available from 4 countries (Northern Ireland, Sweden, Germany and Denmark). Patients from Portugal and the Republic of Ireland could not be included as they did not complete the 18-month study: comparative data were not collected from control patients in The Netherlands. At baseline, 41.7% of the intervention patients and 41.3% of the control patients reported one or more hospitalisations during the 18 months prior to the study. During the 18-month study, a lower proportion of intervention patients reported one or more hospitalisations compared with control patients (35.6 and 40.4%, respectively); however, this difference was not statistically significant (Chi-squared test, $p > 0.05$). Individual country analysis showed that in Denmark the difference during the study was significant, with a smaller proportion of intervention patients reporting one or more hospitalisations compared with control patients (31.3 and 42.3%, respectively; Chi-squared test, $p = 0.04$).

#### *Clinical Signs and Symptom Control*

Intervention patients agreed that they controlled their medical conditions better during the study than before participation in the project at the 6-, 12- and 18-month assessment periods (73, 71 and 75% respectively; data on this parameter were not available for The Netherlands).

#### *Patient Satisfaction with Service Provided*

Three measures were used to monitor patient satisfaction with pharmacy services: rating of services provided, overall satisfaction with services and general opinion of pharmaceutical care. Data relating to the latter aspect were only collected from intervention patients.

##### Rating of Services Provided

At baseline, approximately two-thirds of both control and intervention patients rated the services provided by their pharmacy as excellent (66.2 and 68.2%, respectively). Over time, the proportion of intervention patients rating the services as excellent increased, whereas in control patients, the proportion remained relatively constant: the percent-

**TABLE III TO GO HERE**

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ages of intervention and control patients rating the service as excellent at 6 months were 72.8 and 63.7%, respectively; at 12 months, 73.4 and 71.2%, respectively; and at 18 months, 73.8 and 64.6%, respectively. Between-group analysis for the pooled data indicated that intervention patients rated the services significantly higher at 6 and 18 months compared with control patients (Mann-Whitney test, $p < 0.05$), and within-group analysis over time for the pooled data indicated that in the intervention group the increase in the rating was significant between the baseline and the 18-month assessment (Wilcoxon test, $p = 0.018$), whereas the changes within the control group were not.

### Satisfaction with Services

Overall, both intervention and control patients were very satisfied with the services provided, with over 90% of both control and intervention patients agreeing (agree/mainly agree) with the statement 'I am satisfied with the services provided by the pharmacy that I regularly visit' at each assessment period. Very few patients indicated that they were dissatisfied (<1% of patients at each assessment). There were no differences between the control and intervention patients at any assessment point (Mann-Whitney test, $p > 0.05$); however, there were small but statistically significant increases in satisfaction in the intervention group over time (baseline 92.0% *vs* 6 months 95.1%, Wilcoxon test, $p = 0.012$; baseline 92.0% *vs* 12 months 93.9%, Wilcoxon test, $p = 0.039$). No significant change over time was observed for control patients (Wilcoxon test, $p > 0.05$).

### General Opinion of Pharmaceutical Care

Overall, intervention patients had a positive view of the pharmaceutical care programme, with over 75% of patients reporting that it was better than the service received prior to the study (6 months 74.9%; 12 months 81.3%; 18 months 77.0%). The majority of the remainder indicated that the service was neither better nor worse, with only a small number of intervention patients indicating that the pharmaceutical care provided was worse than the care received prior to participation

**Table III.** Change in 36-Item Short Form (SF-36) health survey scores over time using the area under the curve summary measure

| SF-36 domain | Denmark | | Germany | | The Netherlands | | Northern Ireland | | Portugal | | Sweden | | Total | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | I | C | I | C | I | C | I | C | I | C | I | C | I | C |
| General health | +1.84* | –1.55 | +0.51 | –0.04 | +0.44 | –0.16 | –2.56 | –1.36 | +0.78 | +1.15 | –0.87 | –0.80 | +0.28 | –0.66 |
| Mental health | –0.08 | –1.74 | +0.49 | –2.26 | –0.25 | –1.04 | –3.59 | –0.83 | –2.84 | +0.92 | –2.13 | –1.46 | –0.80 | –1.34 |
| Physical functioning | +0.26 | –1.55 | +2.35 | +1.92 | –0.78 | –1.06 | –6.83* | +7.14 | –7.47 | +1.74 | –1.75 | –2.67 | –0.95 | –0.68 |
| Role emotional | +3.45* | –2.73 | +2.94 | +2.57 | +1.02 | +2.52 | –7.33 | –4.46 | +9.26 | –4.04 | –4.56 | –5.13 | +0.20 | –2.88 |
| Role physical | +4.43 | –0.39 | +1.88 | –4.41 | –2.76 | +1.96 | –6.52* | +7.11 | –1.87 | +0.48 | –4.52 | –4.56 | –1.10 | –0.31 |
| Vitality | +0.98 | –1.57 | +1.00 | –3.53 | –1.06 | –0.66 | –2.26* | +7.24 | –1.56 | –1.83 | –1.28 | –1.00 | –0.46 | –0.82 |
| Bodily pain | +0.81 | –0.17 | +2.64 | –1.20 | –0.69 | +1.17 | –2.22 | +3.78 | –0.93 | +2.01 | –1.05 | –0.42 | –0.06 | +0.53 |
| Social functioning | +0.49 | –1.97 | +0.21 | –4.59 | –0.43 | +0.01 | –0.86 | –0.22 | –5.63 | –3.84 | –2.62 | –2.34 | –0.65 | –1.51 |

**C** = control; **I** = intervention; * $p < 0.05$ (independent t-test) *vs* control.

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(6 months, 2 patients; 12 months, 2 patients; 18 months, 3 patients).

### Cost of Healthcare and Associated Healthcare Needs

When each of the component costs were expressed as a percentage of the total cost, the most important components for all countries (and at all 3 assessment periods) were the costs of drugs and hospitalisations. The mean total costs per patient (in EUR) within each country are displayed in table IV. Between-group analysis indicated that there were no significant differences between the total cost for control and intervention patients in any country (Mann-Whitney, $p > 0.05$); however, in some countries, there were significant differences between control and intervention patients in relation to the individual components. For example, in Germany, intervention patients had significantly lower costs associated with hospitalisations and contacts with specialists compared with control patients in the second 6-month period (Mann-Whitney test, $p < 0.05$). Cost savings were achieved in most of the countries at each assessment period (i.e. the average cost per intervention patient was lower than that of the control patients) [table IV].

### Intermediate Outcomes

Table V outlines the results for the intermediate outcomes measured during the study. In the pooled data, there were no significant differences between the control and intervention patients at any assessment point with regard to knowledge of medicines, contact with GPs or prescription and nonprescription drug use. Although similar proportions of intervention and control patients were categorised as being compliant at each assessment point, an analysis of changes in compliance during the study (change in compliance status compared with that reported at baseline) indicated that at 18 months a significantly higher proportion of the intervention patients changed from being noncompliant to compliant compared with the control patients (15.2 and 12.2%, respectively; Chi-squared, $p = 0.028$). In

**Table IV.** Total average cost per patient [in euros (EUR) January 1999] during implemented pharmaceutical care programme within each country

| Country[a] | Total average cost per patient (EUR) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | first assessment period (0-6 months) | | | second assessment period (6-12 months) | | | third assessment period (12-18 months) | | |
| | intervention | control | cost saving | intervention | control | cost saving | intervention | control | cost saving |
| Denmark | 1473.57 | 1836.00 | +362.43 | 1850.86 | 1572.14 | –278.72 | 1298.13 | 1419.88 | +121.75 |
| Germany | 3350.97 | 3424.77 | +73.80 | 2483.19 | 3700.05 | +1216.86 | 2992.25 | 3167.25 | +175.00 |
| Republic of Ireland[b] | 1604.07 | 1236.55 | –367.52 | | | | | | |
| Northern Ireland | 789.10 | 1217.86 | +428.76 | 641.93 | 770.89 | +128.96 | 735.22 | 750.01 | +14.79 |
| Portugal[c] | 1858.43 | 2010.50 | +152.07 | | | | | | |
| Sweden | 1314.05 | 1152.56 | –161.49 | 1016.35 | 1218.64 | +202.29 | 1266.76 | 1250.34 | –16.42 |

a Data from The Netherlands were not analysed as comparative data from control patients were not collected.

b Owing to operational difficulties, only 6 months' data were collected.

c Based on data for a 12-month period (0-12 months).

the pooled data, there were significantly more changes in medication self-reported by the intervention group than by the control group at baseline and the 6-month assessment (Mann-Whitney test, $p < 0.05$). However, although there was some fluctuation in the mean number of self-reported changes in medicines over time, no general pattern was observed for this variable.

In response to a statement relating to recommendations made by pharmacists, approximately 90% of pharmacists agreed that the recommendations that they suggested were welcomed by GPs (6 months, 89%; 18 months, 93%). The responses of GPs to a similar statement indicated that 50% of the GPs at both assessment points agreed that they considered the recommendations suggested by pharmacists when deciding on treatment for the intervention patients. In Denmark, a larger proportion (90%) of GPs gave a positive response to this question. Of the intervention pharmacists, 80% had a positive opinion of pharmaceutical care and its provision (17% neutral, 3% negative); corresponding data for the GPs were 52% positive, 46% neutral and 2% negative. Again, there were differences between the countries, for example, 92% of GPs and 100% of pharmacists in Sweden were positive about pharmaceutical care provision, while in Denmark, the corresponding percentages were 48 and 78%.

## Discussion

The present study adds to the limited evidence-based research evaluating the effectiveness of pharmaceutical care provision via community pharmacists. Although the effects of community pharmacy-based pharmaceutical care programmes aimed at the older patient have been previously assessed,[15-18] the documented studies have involved only small numbers of patients, have investigated limited outcomes and have been criticised for their methodology and study design.[6] The present research is the first large-scale, multicentre, controlled, longitudinal study investigating the effects of pharmaceutical care provision by community pharmacists to elderly patients.

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**Table V.** Summary of data for intermediate outcome measures at each assessment point

| Intermediate outcome | Baseline | | 6 months | | 12 months | | 18 months | |
|---|---|---|---|---|---|---|---|---|
| | I | C | I | C | I | C | I | C |
| Knowledge of medicines[a] (mean score ± SD) | 65.1 ± 14.51 | 63.2 ± 17.0 | +3.85 ± 14.61 | +2.28 ± 16.10 | +3.73 ± 14.75 | +4.46 ± 16.53 | +3.19 ± 15.18 | +3.16 ± 16.19 |
| Self-reported compliance[b] (% of patients compliant) | 33.9 | 38.6 | 38.5 | 36.6 | 43.8 | 37.3 | 38.2 | 39.4 |
| No. of contact with GPs[c] (mean ± SD) | 4.79 ± 8.39 | 4.27 ± 6.17 | 3.97 ± 5.69 | 3.57 ± 4.58 | 4.00 ± 7.00 | 3.53 ± 5.54 | 4.25 ± 7.98 | 3.24 ± 4.03 |
| Prescription drug use (mean no. ± SD) | 7.05 ± 2.51 | 6.97 ± 2.51 | 6.94 ± 2.72 | 7.01 ± 2.68 | 6.96 ± 2.63 | 6.94 ± 2.75 | 7.14 ± 2.90 | 7.00 ± 2.74 |
| Nonprescription drug use (% of patients reporting using) | 52.0 | 52.1 | 54.1 | 53.2 | 53.3 | 49.4 | 56.2 | 52.8 |
| No. of changes in therapy (mean ± SD) | 1.09* ± 1.32[d] | 0.85 ± 1.18[d] | 1.47* ± 1.80 | 1.09 ± 1.38 | 1.27 ± 1.58 | 1.21 ± 1.49 | 1.37 ± 1.51 | 1.40 ± 1.43 |

a Values for 6, 12 and 18 months are differences in scores attained between assessment period and baseline.
b Data from The Netherlands not included as questions not administered at baseline.
c Included home visits and surgery appointments.
d Changes in medication compared with that given in the 6 months prior to the start of the study.

**C** = control; **GPs** = general practitioners; **I** = intervention; **SD** = standard deviation; * $p < 0.05$ (Mann-Whitney test) *vs* control.

The results indicate that the pharmaceutical care intervention programme had some positive effects on humanistic health outcomes such as satisfaction with treatment, and sign and symptom control, and on economic outcomes, but had less impact than anticipated on drug therapy, drug knowledge and compliance with medication. Overall, the results indicated that there was a general trend for both intervention and control patients' health-related quality of life to remain the same or decline during the period of the study; however, in Denmark, significant improvements in some domains of the SF-36 health survey were demonstrated in intervention patients, whilst there was a decline in the control patients.

A recent review reported that pharmacists' interventions inconsistently demonstrate positive effects on health-related quality of life.[40] The inability of the pharmaceutical care programme provided in this study to significantly affect health-related quality of life in the elderly may be related to several factors. Within this patient population, this may partially be due to a greater disease burden experienced by the elderly and, hence, the use of medication may only have had a small effect on quality of life.[13] Furthermore, despite cross-sectional studies suggesting that health-related quality of life decreases with age,[36,41] there is a lack of research on longitudinal changes in health-related quality of life using the SF-36 health survey.[42]

Despite there being only a slight decline in the percentage of intervention patients reporting hospitalisations, this reduction is encouraging since a reduction in hospitalisations, in particular drug-related admissions, is an important final outcome of pharmaceutical care because the latter has been shown to be the major component of the cost associated with drug-related morbidity and mortality.[43] However, detailed data indicating the reason for admission were not documented during the study, hence the reduction observed could not be attributed to the prevention of drug-related hospitalisation in the intervention patients.

A large proportion of the intervention patients perceived that they had more control over their medical conditions during the study than they did prior to involvement. This is encouraging as it provides further evidence of the benefit of such a model of practice in ensuring patients become more involved in self-care and self-monitoring of their health.[8,9]

A high proportion of all of the patients were satisfied with the services provided by their community pharmacist throughout the study. Despite this ceiling effect and the limitations of using single item measures, there were significant improvements in levels of satisfaction in intervention patients throughout the study period. Intervention patients also rated the services received significantly higher than did the control patients. In future studies, it may be more appropriate to relate patients' satisfaction to their expectations of pharmacy services and, in addition, use a scale that contains a number of items, each assessing a different aspect of satisfaction with services.[44-46] Over 75% of the intervention patients liked the new approach taken by the pharmacist. Although the proportion was lower than that of 95%, reported by Burgess and White,[16] the overall positive responses to the services provided by community pharmacists as part of pharmaceutical care are encouraging and indicate that such a service is welcomed by patients. However, the positive response may have been due to the increased social interaction with the pharmacist rather than an actual judgement of the technical quality of the care that was provided.[47]

In most of the participating countries, cost savings were observed in intervention patients compared with control patients. Many of these cost savings were as a direct result of reduced drug costs or reduced costs associated with hospitalisations. These results support previous research that has documented reduced costs following pharmaceutical care provision.[9,12,48,49] In order for such a model to be adopted by community pharmacists and offered routinely to high risk elderly patients, sufficient remuneration for this extended service must be secured. Implementing such a pharmaceutical care programme within primary care would incur considerable costs;[50] however, these could be offset against the expected savings.

The pharmaceutical care programme that was implemented did not appear to have any effect on medication knowledge, usage of medicines (compliance and number of medicines used) or contact with GPs. Other researchers have also reported that pharmacist intervention had no effect on these parameters;[13,15] however, there are some reports that have documented an improvement in knowledge of medicines,[51,52] compliance[53] and physician visits[48] following increased pharmacist involvement in care provision. A limitation of the results obtained in the present study is that the measurement tools used to measure these outcomes relied on self-reports by the patient. In future, the instruments used to measure knowledge should assess more complex aspects of knowledge such as comprehension and understanding of the drug regimen, rather than checking basic items of knowledge, and pharmacy refill records should perhaps be considered for measuring compliance as these have been shown to provide a valid, objective measure of compliance.[54]

In addition to these quantitative methods of measuring outcomes, qualitative research methodologies, such as focus groups, could also be employed in future research. Qualitative research may be of particular value in investigating aspects of satisfaction and perceptions of the service both from the perspective of the patient and the healthcare professional. Care should, however, be taken to avoid assessment overload, which could have a negative impact on continued participation of participants in a study of this nature.

The pharmacists involved in providing pharmaceutical care had a positive opinion on the new approach as did the majority of GPs. A limitation of these results is the poor response rates from GPs in a number of countries; this may be a reflection of the limited contact that occurred between intervention pharmacists and GPs during the study, despite encouragement from the research centres. This highlights the disjointed nature of primary care in many countries – a collaborative approach is needed for the future provision of pharmaceutical care, where GPs and pharmacists are involved in joint education programmes to facilitate better contact between the professions and to develop team-work within primary care.[55,56]

This multicentre study was the first of its kind in the field and, thus, it is not surprising that there were a number of methodological issues that affected the implementation of the study. A number of these issues will be of interest to other researchers planning to undertake community pharmacy-based studies, and they are discussed below. Although a substantial number of patients in total were recruited into the study by a large number of community pharmacists, lower than anticipated numbers were recruited in some countries (Northern Ireland, Republic of Ireland and Portugal). Lack of interest by patients in being involved in the study was also reported by a number of pharmacy sites, which may be due to the novel situation of such a trial being performed in a community pharmacy setting and the added difficulty of recruiting the elderly.[57] It has been suggested that pharmaceutical care providers may be able to improve participation in future studies by informing and educating patients about the potential benefits of such a service.[58] Formal involvement of GPs from the outset may also improve patient recruitment and aid in targeting those elderly patients who would benefit most from the service.

Patient withdrawal was substantial during the study. Despite patients who were lost to follow-up being older and reporting poorer quality of life compared with those who completed the study, the profiles of control and intervention patients who dropped out were otherwise similar. The main reason for patients being lost to follow-up was unwillingness to continue with the study. In a number of countries, withdrawal of pharmacy sites also contributed significantly to patients being lost to follow-up. The unwillingness of patients to continue with the study could be explained, in part, by the burden placed on them to complete study questionnaires and, again, by unfamiliarity with research of this nature being performed with the community pharmacy. Furthermore, it was felt by the pharmacists involved that those patients who agreed to participate in the study were those who already had a good relationship with their pharmacists, who were knowledgeable about their medicines and who managed their chronic disease states well. This has been reported by other pharmacy practice researchers,[15] and, as such, it minimises the possible impact that can be achieved under study conditions.

One aspect of the study that was not rigorously controlled was the training of participating pharmacists. A study manual was provided to each participating pharmacist, followed by a 1-day training session. Further training was provided in individual countries; however, the extent of this was driven by the available resources. In retrospect, a standardised training package, in addition to the study manual, should have been developed and used to formalise the type and extent of training received. It was generally felt by the researchers that additional training of pharmacists, and indeed professional training involving the GPs of intervention patients, could have had a positive impact on study outcomes. Furthermore, it is clear that rigorous monitoring of process indicators is required to ensure that the proposed intervention programme is being implemented.

The enthusiasm of the current investigators to evaluate the effects of pharmaceutical care on a large number of outcomes may have adversely affected the data quality and participation in the project by those involved. Therefore, future projects of this nature should adopt a more streamlined approach to data collection. However, there is still a great need for succinct, validated instruments suitable for measuring the outcomes of pharmaceutical care. The Pharmaceutical Care Network in Europe,[59] which includes researchers from all of the participating centres, recently addressed this issue while holding a 3-day workshop aimed at developing outcome measures for use in future pharmaceutical care research projects.[60] Where possible, objective measures (e.g. data on the use of healthcare resources) should be obtained from medical records rather than by depending on self-report by patients, especially in the case of elderly patients whose cognitive function may be in decline.

Barriers to the successful implementation of pharmaceutical care have been documented previously.[61,62] Similar barriers were identified in the present study, for example, lack of time to devote to data collection and implementation of the study or higher priority activities to be performed within the pharmacy.[63] It was the collective view of all of the participating pharmacists and researchers that the full potential of pharmaceutical care provision by community pharmacists to elderly patients was not realised during the study and that the patient group-specific model used did not appear to be as effective as the disease-specific models that have been researched.[8-11,52]

## Conclusion

The present study was the first large-scale, multicentre, controlled, longitudinal study investigating the effects of pharmaceutical care provision by community pharmacists to elderly patients and, as such, it represents a major advance in knowledge and experience of performing research of this nature in the primary care setting. The positive effects observed during the study appear to have been achieved via the social and psychosocial aspects of the study such as the increased support provided by the community pharmacist and the improved patient-pharmacist relationship, rather than via biomedical mechanisms. Considerable knowledge and experience was gained by the researchers during the investigation and the following general recommendations are made for future research into the pharmaceutical care of elderly patients within the primary care sector:

- Improve patient targeting so that those patients in greatest need are provided with pharmaceutical care.
- Encourage a higher proportion of these latter patients to participate in primary care-based research projects.

- Modify and simplify outcome measure assessments using reliable and valid measurement tools where available.
- Quantitative measurements should be supplemented by qualitative methodology. Future research may investigate the impact of psychosocial factors on healthcare outcomes.
- Ensure a multidisciplinary approach to pharmaceutical care provision, with pharmacists and other primary healthcare practitioners (i.e. GPs and nurses) working more closely together.

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